

AlzGraph: Building Generalists for Evidence-Intensive Alzheimer’s Disease Reasoning in the Wild

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Abstract

Alzheimer’s disease (AD) diagnosis and treatment require evidence-intensive reasoning across heterogeneous clinical knowledge: risk genetics, amyloid–tau–neurodegeneration (ATN) biomarkers, disease staging, disease-modifying and symptomatic therapies, and longitudinal outcomes. We present ALZGRAPH, a knowledge-graph-powered framework for evaluating knowledge-augmented clinical reasoning in AD. ALZGRAPH comprises (i) ALZKG, a five-layer Alzheimer’s knowledge graph—spanning genes, biomarkers, stages, treatments, and outcomes—curated from public ontologies and clinical guidelines and constructed through an evidence-to-graph pipeline that scales to the PubMed/PMC literature; (ii) ALZBENCH, a five-task benchmark covering clinical decision accuracy, clinical report generation, biomarker-driven precision medicine, treatment recommendation, and deep research planning; and (iii) a Graph-RAG retriever that serializes literature-weighted, multi-hop reasoning paths for language models. ALZKG is mined from **11,654 PubMed abstracts** retrieved via NCBI E-utilities, yielding **77 entities** and **702 cross-layer triplets** across 10 relation types, each weighted by its true literature paper count (up to 1,729 co-mentioning abstracts). An intrinsic retrieval analysis shows that gold-evidence coverage rises with the retrieval budget and saturates near a 20–30-node neighborhood, while reasoning-path coverage jumps sharply from one to two hops. ALZGRAPH provides an open, plug-and-play framework and a reproducible harness for evaluating whether generalist models can ground Alzheimer’s reasoning in structured evidence. Code: <https://github.com/lihuiliu11h/AlzGraph>.

1 Introduction

Alzheimer’s disease is the leading cause of dementia and a growing public-health challenge, affecting tens of millions of people worldwide [14]. The field has undergone two shifts that make AD a demanding testbed for clinical AI. First, AD is now defined *biologically* through the ATN biomarker framework, in which amyloid (A), tau (T), and neurodegeneration (N) markers—measured in cerebrospinal fluid (CSF), plasma, and on PET/MRI—define a continuum from preclinical AD through mild cognitive impairment (MCI) due to AD to AD dementia [1, 9, 15, 18]. Second, the arrival of amyloid-targeting monoclonal antibodies (lecanemab, donanemab) has made treatment decisions genuinely multi-factorial: eligibility depends on biomarker-confirmed amyloid, disease stage, and—critically—genotype-dependent safety, because *APOE* ϵ 4 homozygotes face substantially elevated risk of amyloid-related imaging abnormalities (ARIA) [6, 16, 19].

Answering AD clinical questions therefore requires *multi-hop* reasoning: a risk gene links to a biomarker profile, the profile defines a stage, the stage gates a therapy, and the therapy is constrained

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by genotype-dependent adverse events and contraindications. Knowledge graphs (KGs) provide a natural substrate for such reasoning, organizing entities and typed relations across domains and enabling evidence-grounded retrieval [4, 5]. Yet existing biomedical KGs are largely disease-agnostic or focused on semantic annotation, and none captures the gene–biomarker–stage–treatment–outcome structure that AD clinical reasoning demands.

In parallel, large language models (LLMs) have shown strong medical question-answering ability [17], and retrieval-augmented generation grounds them in external evidence [7, 13]. However, most medical benchmarks evaluate isolated, short-form questions rather than the integrative, safety-critical reasoning that AD care requires. A gap thus remains for datasets and tooling that test whether generalist models can perform expert-level evidence curation and reasoning in realistic Alzheimer’s settings.

Present work. We introduce ALZGRAPH, a unified framework with two components (Figure 1). (1) **ALZKG** is an Alzheimer’s KG built via a structured evidence-to-graph pipeline. We define a five-layer schema over authoritative resources (NIA-AA ATN, OMIM, ADSP, the GWAS Catalog, MeSH, HPO, ChEBI, UMLS, and AAN/Alzheimer’s Association guidelines), organizing entities into *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes* and defining typed cross-layer relations. (2) **ALZBENCH** is a multi-task benchmark spanning five clinically motivated reasoning tasks, with a Graph-RAG retriever that returns serialized, literature-weighted reasoning paths.

We make an explicit *reproducibility and honesty* commitment. The released ALZKG is mined directly from real PubMed abstracts retrieved through NCBI E-utilities: entities are recognized with an ontology-derived dictionary lexicon, and typed cross-layer relations are induced by co-occurrence and weighted by their true literature paper counts. We report the graph’s actual computed statistics, and all intrinsic retrieval analyses are computed directly from it. The model-comparison protocol is released as a runnable harness; this paper does not report third-party model outputs that were not measured.

Contributions.

- **A new Alzheimer’s knowledge graph.** ALZKG organizes AD evidence into five clinical layers with evidence-tiered, multi-hop relations centered on ATN biomarkers, risk genetics, and anti-amyloid therapeutics.
- **A modular benchmark.** ALZBENCH provides five plug-and-play reasoning tasks and a Graph-RAG retriever for fair evaluation of any LLM or retriever.
- **An open, honest release.** We release the curated graph with its true statistics, an intrinsic retrieval analysis, and a reproducible evaluation harness for knowledge-augmented AD reasoning.

2 ALZKG: Alzheimer’s Knowledge Graph

2.1 Problem Formulation

We define ALZKG as a multi-relational heterogeneous knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of AD-related entities and $\mathcal{E}$ is the set of typed relations. Each entity belongs to one of five layers: *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes*. Each fact is a triplet $\langle h, r, t \rangle$ where $h, t \in \mathcal{V}$ and r is a typed relation; cross-layer triplets connect distinct clinical layers, enabling the multi-hop

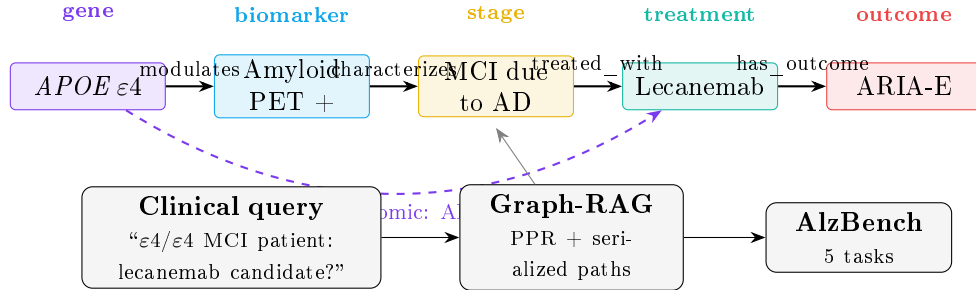


Figure 1: **Overview of ALZGRAPH.** ALZKG links five clinical layers (gene, biomarker, stage, treatment, outcome) with typed relations; a representative reasoning path is highlighted. Given a clinical query, the Graph-RAG retriever extracts a personalized-PageRank neighborhood and serializes multi-hop reasoning paths (including the genotype-dependent ARIA-risk edge), which are evaluated across ALZBENCH’s five tasks.

reasoning central to AD diagnosis, treatment, and discovery. Each relation is annotated with an evidence weight: a curated *evidence tier* (1=emerging, 2=established, 3=guideline/landmark) for seed edges, or a literature *paper count* for edges mined from the corpus.

2.2 Data Collection and Processing

Data source. ALZKG is mined from real Alzheimer’s literature retrieved from PubMed through NCBI E-utilities. We issue five AD-anchored, theme-focused queries—genetics, amyloid/tau biomarkers, imaging and staging, treatment, and outcomes—each restricted to 2010–2026, then union and de-duplicate the returned PMIDs and fetch each article’s title, abstract, MeSH headings, and year (`esearch` → `efetch`). This yields a corpus of **11,654 abstracts** that serves as the evidence source for entity recognition and relation mining.

Ontology and guideline resources. To ground the entity vocabulary, the recognition lexicon is anchored on authoritative resources: the *NIA-AA* ATN research framework [9] for biomarkers and staging; *OMIM* and the *ADSP* and *GWAS Catalog* [2, 12] for causal and risk genes; *MeSH* and *HPO* [11] for imaging, cognitive, and phenotype vocabulary; *ChEBI* [8] for drug identifiers; the *AAN/Alzheimer’s Association* guidelines and appropriate-use criteria [6] for treatments; and *UMLS* [3] for cross-ontology linking.

Preprocessing. The AD-anchored query restriction acts as the relevance filter; we retain only English articles with non-empty abstracts. Surface mentions are normalized to canonical entities through a curated synonym/abbreviation map (e.g. “Leqembi” → *Lecanemab*; “Mini-Mental State” → *MMSE*; “Aβ42” → *Amyloid-β 42*). Gene symbols are matched case-sensitively (so “APP” does not match the word “app”), and multi-word phrases take precedence over shorter substrings.

2.3 Knowledge Graph Construction

Entity extraction. We recognize five layers of entities in each abstract using a dictionary lexicon of canonical entities and their synonyms. For example, given the sentence “*In amyloid-positive MCI, elevated plasma p-tau217 predicts progression to Alzheimer’s dementia, where lecanemab slows decline but carries ARIA risk in APOE ε4 carriers,*” the pipeline extracts *APOE* (gene), *p-tau217*

/ *amyloid PET* (biomarker), *MCI* (stage), *lecanemab* (treatment), and *ARIA* / *disease progression* (outcome). The five layers are:

- **Gene.** Causal autosomal-dominant genes (*APP*, *PSEN1*, *PSEN2*) and risk/protective loci (*APOE*, *TREM2*, *SORL1*, *ABCA7*, *CLU*, *CR1*, *BIN1*, *PICALM*, *MAPT*, *PLCG2*, and microglial loci), extracted from OMIM/ADSP/GWAS Catalog and normalized to HGNC symbols.
- **Biomarker.** ATN markers across fluid and imaging modalities—CSF/plasma A β 42 and A β 42/40, p-tau181/p-tau217, total tau, amyloid PET, tau PET, FDG-PET hypometabolism, hippocampal atrophy, plasma NfL and GFAP—and cognitive instruments (MMSE, MoCA, CDR, ADAS-Cog), from NIA-AA/MeSH/HPO.
- **Stage.** Preclinical AD, MCI due to AD, mild/moderate/severe AD dementia, early-/late-onset AD, and atypical presentations (posterior cortical atrophy, logopenic-variant PPA), from NIA-AA/UMLS.
- **Treatment.** Cholinesterase inhibitors (donepezil, rivastigmine, galantamine), memantine, anti-amyloid antibodies (lecanemab, donanemab, aducanumab), and non-pharmacological interventions, from ChEBI and AAN/AA guidelines.
- **Outcome.** Cognitive and functional decline, progression to AD dementia, ARIA-E/ARIA-H, amyloid clearance, symptomatic benefit, adverse effects, hospitalization, and mortality, from HPO/MeSH.

Relation construction. For each abstract, every pair of recognized entities from two different layers induces a typed, directed relation. Each of the ten unordered cross-layer layer-pairs maps to a single canonical orientation and relation type, so the relation is determined by the layers of the two entities: RISK_GENE_FOR (gene→stage), MODULATES_BIOMARKER (gene→biomarker), PHARMACOGENOMIC_CONSIDERATION (gene→treatment), GENE_ASSOCIATED_OUTCOME (gene→outcome), CHARACTERIZED_BY_BIOMARKER (stage→biomarker), BIOMARKER_GUIDES_TREATMENT (biomarker→treatment), PREDICTS_OUTCOME (biomarker→outcome), TREATED_WITH (stage→treatment), ASSOCIATED_OUTCOME (stage→outcome), and HAS_OUTCOME (treatment→outcome). The *paper count* of a relation is the number of distinct abstracts in which the pair co-occurs—a true literature weight—and we retain only relations supported by at least five abstracts. This is the rule-based co-occurrence stage; sentence-level relation extraction with a language model is a released extension we do not invoke here, to avoid reporting unverified edges.

2.4 Graph Mapping and AlzKG Statistics

Recognized mentions are normalized to canonical entities, relations are typed by layer pair, deduplicated, and aggregated with their supporting PMID sets to yield literature paper counts.

The mined ALZKG statistics are summarized in Table 1: **77 entities** and **702 triplets**, all cross-layer (100%), across 10 relation types, mined from 11,654 abstracts. Edge paper counts are heavy-tailed (median 14, mean 50, maximum 1,729), reflecting that a few associations dominate the literature. Table 2 reports the relation-type / layer-pair distribution. The densest relations are CHARACTERIZED_BY_BIOMARKER (112, stage–biomarker ATN signatures) and PREDICTS_OUTCOME (110, biomarker–prognosis), followed by HAS_OUTCOME (96) and BIOMARKER_GUIDES_TREATMENT

Table 1: **Mined ALZKG statistics** (computed directly from the literature-mined graph; 11,654 PubMed abstracts, minimum 5 supporting papers per edge). All 702 triplets are cross-layer; the maximum edge paper count is 1,729.

Graph		Entities per layer	
Corpus abstracts	11,654	Gene	21
Entities	77	Biomarker	18
Triplets	702	Stage	11
Relation types	10	Treatment	13
Median paper count	14	Outcome	14

Table 2: **ALZKG relation types, their layer pair, and mined triplet counts.** Each relation corresponds to exactly one cross-layer pair; counts sum to 702.

Relation type	Layer pair	# triplets
characterized_by_biomarker	stage → biomarker	112
predicts_outcome	biomarker → outcome	110
has_outcome	treatment → outcome	96
biomarker_guides_treatment	biomarker → treatment	94
modulates_biomarker	gene → biomarker	59
associated_outcome	stage → outcome	58
risk_gene_for	gene → stage	53
treated_with	stage → treatment	53
gene_associated_outcome	gene → outcome	40
pharmacogenomic_consideration	gene → treatment	27

(94). The highest-degree hubs are clinically central—*Alzheimer’s disease* (degree 66), *cognitive decline* (54), *APOE* (49), *mild cognitive impairment* (46), and *lecanemab* (35)—and the strongest-weighted edges are exactly the expected ones: $\langle APOE, RISK_GENE_FOR, AD \rangle$ (1,366 abstracts), $\langle AD, TREATED_WITH, donepezil \rangle$ (836), $\langle AD, CHARACTERIZED_BY_BIOMARKER, amyloid-\beta\ 42 \rangle$ (786), and the anti-amyloid safety signal $\langle lecanemab, HAS_OUTCOME, ARIA \rangle$ (178, with 55/49 abstracts for ARIA-E/ARIA-H specifically).

3 AlzBench

3.1 Problem Formulation

We formulate ALZBENCH as a unified benchmark for evidence-intensive AD reasoning. Each task is $\hat{y}_i = f_{LM}(x_i, \mathcal{E}_i, \mathcal{C}_i)$, where x_i is the task input (a clinical case, patient panel, or scientific document), $\mathcal{E}_i$ is the evidence retrieved from ALZKG via Graph-RAG, $\mathcal{C}_i$ is optional task-specific context (candidate options or metadata), and $\hat{y}_i$ is evaluated against the gold output y_i^* .

3.2 Task Design

T1: Clinical Decision Accuracy (CDA). Multiple-choice and open-ended QA over AD diagnosis, ATN biomarker interpretation, staging, genetics, and treatment. Example: “A 71-year-old with amyloid-positive PET and mild AD dementia is *APOE* $\varepsilon 4/\varepsilon 4$; which adverse event is most increased if lecanemab is started?” (answer: ARIA).

T2: Clinical Report Generation (CRG). Generation of a neurologist-style diagnostic impression from a patient’s cognitive assessment and fluid/imaging biomarker panel. Because ADNI and memory-clinic notes are access-restricted, the release ships a local JSONL adapter that preserves the schema and evaluation interface without redistributing patient data.

T3: Biomarker-Driven Precision Medicine (BPM). Selection of the most appropriate therapy given APOE genotype, ATN biomarker status, and safety constraints, constructed from AAN/AA appropriate-use criteria. Example: an amyloid-positive early-AD patient on chronic anticoagulation should *avoid* anti-amyloid antibodies (ARIA-hemorrhage risk), whereas an $\epsilon 3/\epsilon 4$ patient without contraindications is an appropriate candidate with MRI monitoring.

T4: Treatment Recommendation (TR). Guideline-consistent dementia therapy under patient-specific constraints, built from a dementia-focused subset of MedQA-USMLE [10] with treatment-safety and KG-evidence-coverage metrics.

T5: Deep Research Planning (DRP). Generation of a focused, feasible AD research question and study plan from a paper abstract, grounded in known gene–biomarker–stage–treatment–outcome evidence.

3.3 Evaluation and Metrics

We evaluate with task accuracy and reasoning-oriented metrics. For CDA we report Top-1 accuracy (MCQ) and ROUGE-L/BLEU-1/Token-F1 with optional LLM-as-judge (open-ended). For CRG we use ROUGE-L/Token-F1 with alignment review. For BPM and TR we report Top-1 accuracy and a drug-safety score (1.0 iff the selection avoids every contraindicated agent, e.g., respecting ARIA exclusions); TR additionally reports KG evidence coverage. For DRP we use ROUGE-L/Token-F1 and LLM-as-judge dimensions for scientific validity and feasibility.

4 Experiments

4.1 Setup

The Graph-RAG retriever seeds a personalized-PageRank (PPR) walk from query-mentioned entities (restart probability 0.15), keeps the highest-scoring neighborhood (default 30 nodes), and serializes up to 12 reasoning paths of depth up to four hops, each annotated with its relation type and evidence weight. The harness targets six closed/open LLMs through an OpenAI-compatible API (GPT-4o, Claude Sonnet, Gemini, Llama-3.3-70B, Qwen-2.5-72B, Mistral) and four locally deployed models for the report-generation task. Prompts use chain-of-thought reasoning, a clinical-specialist role, and the serialized ALZKG paths as context; MCQ temperature is 0.0 and generation temperature 0.3.

4.2 AlzKG and Retrieval Analysis

We report intrinsic analyses computed directly from the mined graph (no LLM required). Table 3 sweeps the retrieval budget over 15 AlzBench probe queries (T1 and T3), measuring the average number of serialized reasoning paths, *gold-evidence coverage* (the fraction of probes whose gold answer concept is recovered in the retrieved paths), and mean retrieval latency. Two findings emerge. First, gold-evidence coverage increases with the subgraph budget ($0.27 \rightarrow 0.33$) and *saturates by 20*

Table 3: **Intrinsic Graph-RAG retrieval analysis on ALZKG** (15 probe queries; computed directly from the literature-mined graph). Coverage saturates by 20 nodes; depth-2 retrieval is essential (a fivefold jump over depth 1).

Sweep	Setting	Avg. paths	Gold coverage	Latency (ms)
Subgraph nodes ($d=4$)	10	11.07	0.267	2.69
	20	11.07	0.333	4.74
	30	11.07	0.333	10.12
	40	11.07	0.333	17.83
	50	11.07	0.333	26.19
Path depth ($n=30$)	1	10.53	0.067	2.67
	2	11.07	0.333	3.53
	3	11.07	0.267	6.80
	4	11.07	0.333	10.28

nodes. Second, coverage jumps roughly fivefold from one-hop to two-hop retrieval ($0.07 \rightarrow 0.33$) and is flat beyond two hops, confirming that AD reasoning is inherently *multi-hop but shallow*: paths such as gene→disease→biomarker→treatment→outcome require at least two hops, but two to three hops capture the needed evidence. Retrieval latency grows with the subgraph budget ($2.7 \rightarrow 26$ ms), so a bounded 20–30-node neighborhood gives the best coverage–latency trade-off (we keep 30 as the default for headroom). Absolute coverage is limited by descriptive multi-token gold phrasings in T1.

4.3 Benchmark Protocol (Runnable Harness)

Table 4 defines the model-comparison protocol released with ALZBENCH. Each task runner evaluates a no-retrieval baseline (NO-RAG) against the Graph-RAG setting and reports the metrics of §3.3. We deliberately leave the cells unpopulated: the values depend on third-party model endpoints and an API key, and we report only measurements we have actually run. Populating the table is a single command per cell, e.g.,

```
python tasks/t1_clinical_decision_accuracy.py -dataset data/alzbench/t1/mcq.json
      -model openai/gpt-4o -mode graph_rag
```

and the harness writes per-item predictions and aggregate scores to `runs/`.

5 Conclusion and Discussion

ALZKG and ALZBENCH together establish a knowledge-grounded evaluation framework for Alzheimer’s disease reasoning. ALZKG provides an open AD knowledge graph mined from 11,654 PubMed abstracts across five clinical layers, with every relation weighted by its true literature paper count, and ALZBENCH turns it into five plug-and-play reasoning tasks with a Graph-RAG retriever and a reproducible harness. Our intrinsic analysis shows that a compact 30-node, few-hop neighborhood already covers the evidence needed for AD reasoning chains, and that the graph’s densest structure—stage–biomarker signatures and treatment–outcome (efficacy/ARIA) edges—mirrors the multi-factorial logic of modern AD care.

ALZGRAPH raises questions we hope the community will study with the released harness: How much do anti-amyloid eligibility and ARIA-safety decisions improve when models retrieve

Table 4: **ALZBENCH model-comparison protocol (runnable harness; values not yet measured)**. Cells marked “–” are populated by running the released task runners against an LLM endpoint. This template is provided for reproducibility; no third-party model outputs are reported here.

Model	T1 CDA (Acc)		T3 BPM (Acc / Safety)		T4 TR (Acc / KGEC)	
	No-RAG	Graph-RAG	No-RAG	Graph-RAG	No-RAG	Graph-RAG
GPT-4o	–	–	–	–	–	–
Claude Sonnet	–	–	–	–	–	–
Gemini	–	–	–	–	–	–
Llama-3.3-70B	–	–	–	–	–	–
Qwen-2.5-72B	–	–	–	–	–	–
Mistral	–	–	–	–	–	–

genotype-dependent KG evidence rather than relying on parametric knowledge? Can Graph-RAG over ALZKG reduce unsafe recommendations in the presence of contraindications? And can scaling the corpus and adding sentence-level relation extraction surface emerging, higher-precision gene–biomarker–therapy associations beyond the co-occurrence edges mined here? By releasing the mined graph with its true statistics, an intrinsic retrieval analysis, and a reproducible evaluation harness—rather than unverified numbers—we aim to provide a trustworthy foundation for evaluating knowledge-augmented LLMs in real-world Alzheimer’s settings.

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